Standard Operating Procedure (SOP): OT Network Infrastructure

Document Information

Property	Details
Document Title	Standard Operating Procedure – OT XTRAN Node Connectivity & TXCARE Commissioning
Document ID	SOP-OT-001
Target Audience	IT Engineers, OT Field Engineers, System Administrators
Scope	Operational Technology (OT) Network Infrastructure

1. Purpose

This document provides step-by-step instructions for establishing local connectivity to an OT XTRAN node via its Central Management Module (CSM) and performing the initial node commissioning and discovery using TXCARE (TX-CARE Management System).

2. Prerequisites & Equipment

Before commencing the setup procedure, field engineers must ensure the following equipment and parameters are prepared:

- **Hardware**:
  - Field/Configuration Laptop with an Ethernet port (or USB-to-Ethernet adapter).
  - Cat5e/Cat 6 RJ45 patch cable.
- **Software & Drivers:**
  - TXCARE Management System software installed.
  - Windows Command Prompt (CMD) or equivalent network utility.
- **Network Parameters:**
  - Designated Target Node IP Address and Subnet Mask.
  - Reserved IP range for field management devices

3. Physical & IP Configuration Setup

Step 1: Physical Connection

1. Connect one end of the RJ45 Ethernet patch cable to the Ethernet management port of the laptop/PC.
2. Connect the other end of the cable directly to the **XTRAN CSM (Central Services Module)** management port.

Step 2: Local Network Adapter IP Assignment

1. Open **Network Connections** on your configuration laptop/PC.
2. Right-click your Ethernet adapter and select **Properties**.
3. Select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**.
4. Configure the static IP address to fall within the same subnet as the target XTRAN CSM node.

Subnet Configuration Example:

Parameter	Configuration Value
Target XTRAN Node IP	192.0.2.97
Subnet Mask	255.255.255.240 (/ 28 network prefix)
Valid Host Address Range	192.0.2.98 to 192.0.2.106
Laptop Assigned IP	192.0.2.98(or any free address within range)

Step 3: Verify Physical & Layer-3 Connectivity

1. Open Command Prompt (cmd).
2. Execute a continuous or 4-packet ping test to verify communication with the CSM node: ping 192.0.2.97
3. Verify that responses are returned with 0% packet loss and low latency before proceeding to software provisioning.

4. TXCARE Installation & Startup

Install **TXCARE** on the configuration laptop/PC and bring it up to a working state (logged in, servers started) before node provisioning in Section 5.

Step 1: Install TXCARE

1. Run the TXCARE installer on the configuration laptop/PC.
2. Follow the installation prompts until the installation completes.

Step 2: Launch TXCARE

1. Open **TXCARE** from the Start menu or desktop shortcut.
2. On the very first start, TXCARE asks for a one-time user database setup (**Create/Select User Database** window). Connect to the MySQL server (Server IP/Host Name 127.0.0.1, Username root) using the MySQL root password, then click **Create/Select**.
3. Wait for the TXCARE Dashboard to appear. If TXCARE fails to open, go to Step 3. Otherwise continue to Step 4.

Step 3: Restart the SQL Server Service (only if TXCARE fails to open)

1. Open **Windows Services**: press **Win + R**, type services.msc and press Enter.
2. In the list, find the **SQL server** service used by the TXCARE database. The OTN manual lists the TXCARE user database as MySQL, so use the exact service name shown on the laptop.
3. Right-click the service and select **Stop**. Wait until the service has fully stopped.
4. Right-click the service again and select **Start**. Confirm the status shows **Running**.
5. Launch TXCARE again and confirm that the Dashboard appears.

Step 4: Log In to TXCARE

1. On the Dashboard, log in using the login box (**Log in** button on the left side of the Dashboard).
2. Use the administrator account for first-time use after installation (predefined username admin; password as given in the vendor user management manual), or the account assigned for the project.

3. Confirm the logged-in username is shown on the Users tile and that the tiles you need are unlocked.

Note: A local user is blocked after 5 wrong password attempts. A blocked user can only be unblocked by a user administrator.

Step 5: Start the TXCARE Servers

1. On the Dashboard, open the **Servers** tile and start the TXCARE servers.
2. Confirm the Servers tile shows **Started**. Some tiles, such as **Discovery**, stay locked while the servers are stopped.
3. Proceed to Section 5.

5. TXCARE Provisioning & Node Commissioning

Perform the following operational steps within **TXCARE** management interface:

Step#	Action Phase	Detailed Description
5.1	**Discovery Entry**	Launch TXCARE and navigate to **Discovery Entry** to initiate node detection on the local subnet.
5.2	**Connection Verification**	Select **Check Connection** to confirm communication stability between TXCARE and the node.
5.3	**Node Approval**	Verify the detected MAC/IP address against project assignment records, then select **Approve Node**.
5.4	**Device Auto-Creation**	Click **Auto Create Devices** to allow TXCARE to enumerate and populate the local hardware architecture.
5.5	**Link Creation**	Select **Create Link** to define and establish logical/physical links for the hardware
5.6	**Device Creation**	Select **Create Device** to finalize the device instances within the TXCARE topology hierarchy.
5.7	**Device Connection**	Click **Connect Device** to initiate active management control.
5.8	**Load Configuration**	Select **Load Config** to deploy baseline operational parameters to the hardware.
5.9	**Error Resolution**	If configuration errors occur, execute **Clear Device Config** and reload the valid configuration file.
5.10	**Ring Topology Creation**	Navigate to topology settings and select **Create Ring** (if building or expanding a resilient ring network architecture).
5.11	**Port Mode Provisioning**	**Critical:** Prior to configuring services, set the targeted port mode to **LAN**.

6. Verification & Acceptance Criteria

- [ ] Node status displays **Active/ Online** with no persistent alarms in TXCARE.
- [ ] IP connectivity remains stable under continuous monitoring.
- [ ] Port modes are correctly mapped to **LAN** prior to service deployment.
- [ ] Topology ring status shows healthy/closed redundancy (if applicable).

7. Document Revision History

Version	Date	Author	Description of Changes
1.0	September 2026	Muhammad Akmal Iman Bin Rosli (Protégé IT Engineer)	Initial draft for XTRAN Connectivity and TXCARE Setup guide.
1.1	November 2026	Muhammad Akmal Iman Bin Rosli (Protégé IT Engineer)	Added Section 4 TXCARE Installation & Startup (install, first-start user database setup, SQL service restart if TXCARE fails to open, login, start servers). Renumbered previous Sections 4 to 6 as Sections 5 to 7.